DO NOW · AO1 REVISION

Electromagnetism

Magnetism and electromagnetism
Paper 2 · Combined and Separate
12 questions

Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know. Learn the 5 starred (★) questions first.

	QUESTION	ANSWER	✓
1	**State when a wire has a magnetic field around it.**	When it carries a current.	☐
2 ★	**Describe the magnetic field around a straight current-carrying wire.**	Concentric circles around the wire.	☐
3	**State what the strength of the magnetic field around a wire depends on.**	The size of the current and the distance from the wire	☐
4	**What is a solenoid?**	A long coil of wire.	☐
5 ★	**Describe the magnetic field around a solenoid.**	Similar to that of a bar magnet.	☐
6	**State where the magnetic field of a solenoid is strong and uniform.**	Inside the solenoid.	☐
7	**Describe how to make an electromagnet.**	Wrap a coil of wire around an iron core and pass a current through it.	☐
8 ★	**Give two advantages of an electromagnet over a permanent magnet.**	It can be switched on and off, and its strength can be varied.	☐
9 ★	**Give two ways to make an electromagnet stronger.**	Increase the current or add more turns.	☐
10	**State the effect of adding an iron core to a solenoid.**	It makes the magnetic field stronger.	☐
11 ★	**State how to reverse the magnetic field around a wire.**	Reverse the current.	☐
12	**A wire is coiled into a solenoid. State the effect on its magnetic field.**	It makes the magnetic field stronger.	☐

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